SAMPLE PROBLEM STATEMENT

Smart Waste Management and Collection System

Reference format: CSA10 – CO1 AT1 Case Study Analysis

1. Industry Context

Cities and towns generate large quantities of solid waste every day. In many places, waste collection follows fixed schedules without considering the actual amount of waste present in each bin. As a result, some bins overflow before collection, while other bins are collected when they are not yet full. These problems can cause unpleasant odours, unhygienic surroundings, unnecessary vehicle trips, increased fuel consumption, and higher operating costs. Manual inspection also makes it difficult for authorities to monitor bins across a large area and respond quickly.

A smart waste management system can address these issues by collecting bin-level information, showing the status and location of bins, and helping collection teams prioritize bins that need service.

2. Problem Statement

Many municipalities and waste management organizations rely on fixed collection schedules and manual inspection to manage public waste bins. This approach does not provide accurate, real-time information about bin capacity, which can lead to overflowing bins, delayed collection, unnecessary collection trips, increased fuel costs, and poor sanitation. The proposed Smart Waste Management and Collection System is a software-based platform that records and monitors waste-bin status, identifies bins that require collection, and helps administrators plan collection activities. The system will allow authorized users to view bin locations and status, receive alerts for bins that are nearly full, maintain collection records, and generate reports. The system should be reliable, secure, easy to use, and capable of supporting additional bins and collection areas as the service grows.

3. Objectives

- Monitor and display the status of waste bins.
- Identify bins that are nearly full and require collection.
- Notify administrators or collection workers when collection is needed.
- Display bin locations to support collection planning.
- Maintain collection schedules and completed-collection records.
- Generate reports about bin status and collection activities.
- Reduce overflowing bins, unnecessary trips, manual work, and operating costs.
- Protect system information through appropriate access controls and security measures.

4. Scope

The first version will focus on registering bins, recording their locations and status, displaying current information on a web interface, notifying authorized users when bins need collection, and maintaining collection records and reports. If IoT sensors are available, they can be used to send bin-level readings automatically. Otherwise, status can be entered manually for the initial prototype. Automatic route optimization, predictive analytics, and large-scale city integration can be considered as future enhancements.

5. Stakeholders and Challenges

Main stakeholders:

- Municipal administrators: manage bins, users, schedules, and reports.
- Waste collection workers: view assigned bins and update collection status.
- Residents and local communities: benefit from cleaner public spaces.
- System developers and maintenance staff: build, secure, and maintain the platform.

Possible challenges:

- Incorrect or delayed bin-status updates.
- Sensor failure or unreliable connectivity when IoT devices are used.
- Training workers and administrators to use the system.
- Protecting user accounts and operational data.
- Maintaining the system as the number of bins and collection areas increases.

6. Proposed Software Engineering Solution

The system can be divided into the following modules:

- User and access management: login and role-based permissions.
- Bin management: register bins, locations, and current status.
- Monitoring and alerts: identify bins that need collection and notify responsible users.
- Collection management: assign collection tasks and record completion.
- Reports: summarize bin status, collection history, and outstanding tasks.

7. Suggested Development Methodology

Agile Scrum is a suitable suggested methodology because the system can be developed in small, testable increments. For example, Sprint 1 can deliver login and bin registration; Sprint 2 can add status monitoring and alerts; and Sprint 3 can add collection records and reports. Feedback from administrators and collection workers can be used to improve each sprint. This methodology is a proposed design choice for this sample, rather than a requirement stated by the reference document.

8. Expected Outcomes and Limitations

Expected outcomes

- Better visibility of bin status and location.
- Faster identification of bins that need collection.
- Fewer overflowing bins and more organized collection work.
- Improved record keeping and reporting.
- Potential reduction in unnecessary trips and operating costs.

Limitations and risks

- Accurate real-time monitoring depends on timely data or functioning sensors.
- IoT hardware and connectivity may increase initial cost and maintenance needs.
- The system supports collection planning but does not guarantee that workers can reach every bin immediately.
- The benefits depend on staff adoption and regular maintenance.

9. Suggested Technologies (Optional)

- Frontend: HTML, CSS, and JavaScript.
- Backend: PHP.
- Database: MySQL.
- Local development environment: XAMPP.
- Optional hardware: ultrasonic sensor and ESP32 for automatic bin-level readings.

Note: The topic and technical choices above are a sample proposal. The CSA10 reference provides the case-study structure and assessment criteria; it does not prescribe this specific waste-management topic or the listed technologies.